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SPMTrack: Spatio-Temporal Parameter-Efficient Fine-Tuning with Mixture of Experts for Scalable Visual Tracking

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Abstract

*Most state-of-the-art trackers adopt one-stream paradigm, using a single Vision Transformer for joint feature extraction and relation modeling of template and search region images. However, relation modeling between different image patches exhibits significant variations. For instance, background regions dominated by target-irrelevant information require reduced attention allocation, while foreground, particularly boundary areas, need to be be emphasized. A single model may not effectively handle all kinds of relation modeling simultaneously. In this paper, we propose a novel tracker called **SPMTrack** based on mixture-of-experts tailored for visual tracking task (TMoE), combining the capability of multiple experts to handle diverse relation modeling more flexibly. Benefiting from TMoE, we extend relation modeling from image pairs to spatio-temporal context, further improving tracking accuracy with minimal increase in model parameters. Moreover, we employ TMoE as a parameter-efficient fine-tuning method, substantially reducing trainable parameters, which enables us to train SPMTrack of varying scales efficiently and preserve the generalization ability of pretrained models to achieve superior performance. We conduct experiments on seven datasets, and experimental results demonstrate that our method significantly outperforms current state-of-the-art trackers. The source code is available at https://github.com/WenRuiCai/SPMTrack.*

1. Introduction

Visual tracking aims to predict the location of a target throughout subsequent frames of a video, given the initial state in the first frame template. Currently, most state-of-the-art trackers adopt a Transformer-based [46] one-stream paradigm [3, 5, 10, 33, 52, 59, 61], utilizing a single Vision Transformer [12] backbone that accepts both the template and the search region image as input, and employs self-attention to perform image feature extraction and template–search region relation modeling simultaneously.

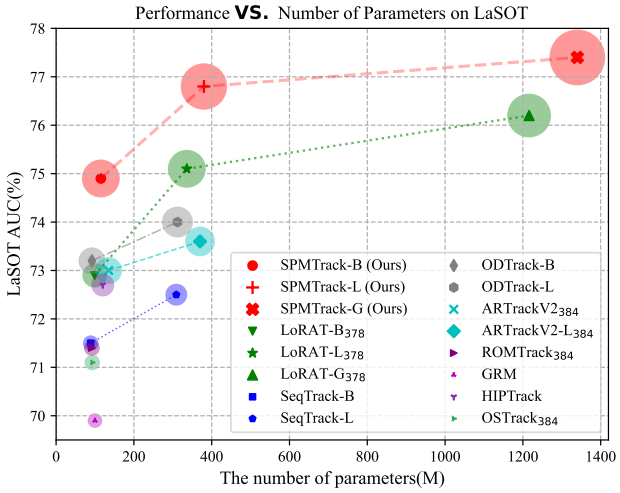


Figure 1. Comparison of LaSOT AUC and model parameter count across different trackers. Larger loop indicates better performance.

However, not all relation modeling between image patches benefits tracking performance. Prior studies [4, 20, 58, 59] have demonstrated that the abundance of background patches in the search region deteriorates the discriminative power of the foreground features, and single vanilla attention has limited capability in suppressing undesirable foreground-background interactions. Meanwhile, there are also some kinds of relation modeling that demonstrate positive effects on tracking performance [17, 19]. For instance, enhancing attention to target boundary regions can improve tracking performance [17], especially in challenging scenarios such as motion blur, partial occlusion, and deformation.

To address various relation modeling, numerous meth-

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ods propose specific designs, including background filtering to reduce target-irrelevant information [3, 59], customizing different interaction modules for tokens belonging to different categories [4, 20], and attention score adjustment based on foreground-background weights of each token [58]. However, existing methods can only handle one or a few predefined types of relation modeling, and the handcrafted strategies inherently lack adaptability. Furthermore, there may exist many latent relationships that further complicate the specialized design. These limitations motivate our question: ***How to design a tracker to adaptively process various relation modeling between image patches?***

Inspired by mixture of experts (MoE) [1, 11, 13, 27, 41] in natural language processing, in this paper, we address the question by proposing **SPMTrack**, a novel tracker based on TMoE, a specialized mixture of experts module for visual tracking task. Similar to MoE, TMoE handles diverse relation modeling through adaptive weighted combinations of multiple experts. However, unlike traditional MoE that is exclusively applied to feed-forward network (FFN) layers, TMoE is simultaneously applied to the linear layers within both the attention layers and the FFN layers. This finer-grained application enables TMoE to achieve more diverse expert combinations, thereby enhancing the capability of various relation modeling. Moreover, differing from traditional MoE that uses FFNs as experts, all the experts in TMoE are linear layers and designed with a lightweight and efficient structure to ensure overall efficiency. Furthermore, we employ TMoE as a parameter-efficient fine-tuning method, which enables us to train SPMTrack of larger scales. In SPMTrack, only a subset of experts and the router within TMoE, along with the prediction head need to be trained, which substantially reduces the trainable parameters while preserving the generalization capability of pretrained models. Notably, our method reduces the number of trainable parameters by nearly 80% compared to [2], while demonstrates better performance.

Benefiting from the powerful capability of TMoE in handling diverse relation modeling, unlike previous one-stream trackers that only perform relation modeling between template-search region pairs, we extend SPMTrack to incorporate multi-frame spatio-temporal context modeling, which further enhances the performance of SPMTrack with minimal additional parameters. We evaluated the performance of our method on seven datasets, and the experimental results demonstrate that SPMTrack achieves state-of-the-art performance across multiple datasets including LaSOT [14], GOT-10K [25], TrackingNet [38] and TNL2K [49]. Through extensive experiments with models of varying scales, as shown in Figure 1, Table 1 and Table 2, with ViT-B [12] as backbone and less than 30% of parameters need to be trained, SPMTrack-B achieves or even surpasses the best performance of previous trackers utilizing ViT-L.

The main contributions of this paper can be summarized as: **(1)** We propose TMoE, a mixture of experts module tailored for visual tracking. TMoE enhances the capability to handle various relation modeling and can be used as a method of parameter-efficient fine-tuning. **(2)** Based on TMoE, we propose SPMTrack, a novel tracker that can effectively integrate spatio-temporal context for visual tracking. **(3)** Experimental results demonstrate that SPMTrack achieves state-of-the-art performance across multiple datasets. We trained SPMTrack of varying scales, and our method with ViT-B as backbone can achieve or even surpass current trackers using ViT-L.

2. Related Work

2.1. One-Stream Trackers

One-stream trackers employ a single Vision Transformer [12] as backbone, simultaneously performing feature extraction and relation modeling at each layer [5, 10, 55, 59]. Many efforts focus on incorporating historical context. TATrack [22] and ODTrack [61] propose architectures capable of processing multiple frames, while HIPTrack [3] introduces historical target features through prompt learning. SeqTrack [6], ARTrack[51], and ARTrackV2 [2] achieve more precise predictions by incorporating multiple historical target position. Other works try to improve conventional self-attention to handle different relation modeling. OSTrack [59] progressively filters background regions to reduce foreground-background relation modeling, GRM [20] and ROMTrack [4] categorizes image tokens and constrains relation modeling between specific categories, F-BDMTrack [58] adjusts the attention score by calculating the foreground-background weights of tokens. But these methods can only handle specific relation modeling and rely on manually designed strategies.

2.2. Mixture of Experts

Mixture-of-experts (MoE) [1, 13, 27] are widely applied in large language models [9, 11, 15, 36, 41, 53], where each expert focuses on specific aspects of the data or particular tasks [48]. By performing an adaptive weighted sum over multiple experts, MoE can better capture complex patterns and relationships within the input data. MoE also has applications in computer vision [8, 43, 60]. However, there is few dedicated exploration of MoE in visual tracking. MoETrack [44] employs multiple prediction heads as experts in the field of RGB-T tracking, which is rather far-fetched and deviates from the general practice of embedding MoE within Transformer layers. In the field of RGB-E tracking, eMoE-Tracker [7] inserts multiple experts between Transformer layers, which more closely resembles an adapter architecture [23] and contrasts with MoE implementations where experts are embedded within Transformer layers.

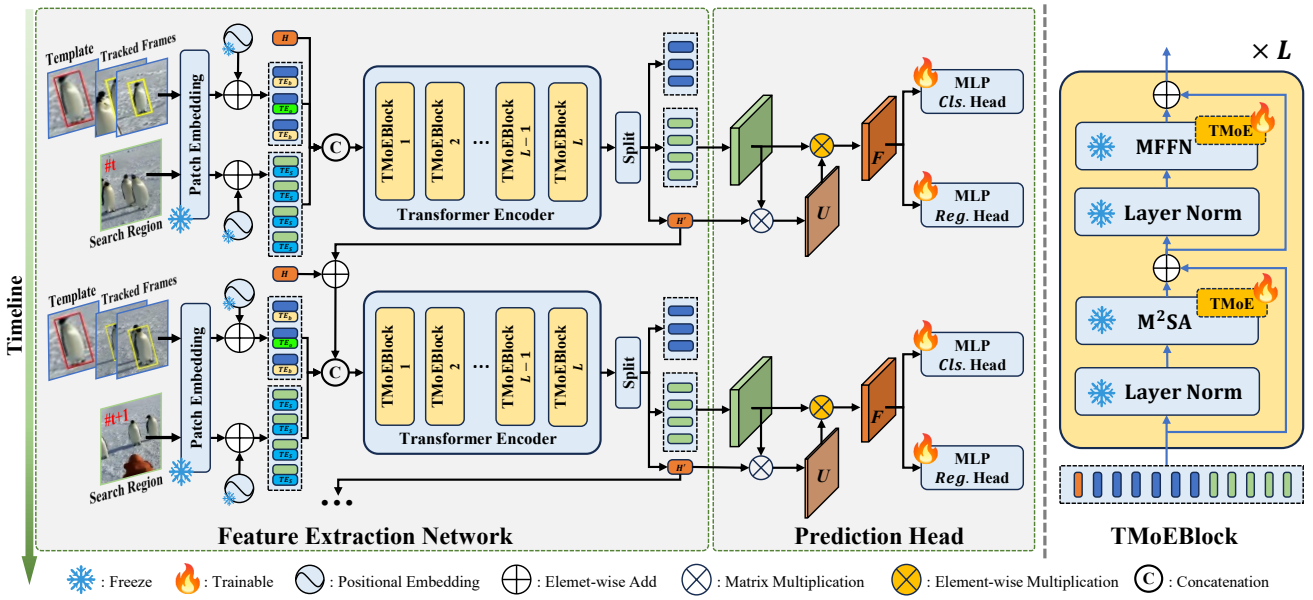


Figure 2. Overview of SPMTrack that consists of a feature extraction network and a prediction head. The main body of the feature extraction network is a Transformer encoder composed of multiple TMoEBlocks. The structure of TMoEBlock is shown on the right side.

2.3. Parameter-efficient Fine-tuning

Parameter-efficient fine-tuning aims to reduce computational costs and retain the general knowledge of the pretrained model by freezing most parameters and fine-tuning only a small subset of parameters. Common parameter-efficient fine-tuning approaches include adapter-based methods [18, 23, 40], prompt-based methods [28, 31, 50, 62] and low-rank adaptation methods [24, 45]. In visual tracking, HIPTrack [3] incorporates historical target features as prompts, while LoRAT [33] employs low-rank adaptation for efficient training. And our method maintains flexibility in diverse relation modeling while reducing trainable parameters by fine-tuning TMoE.

3. Method

3.1. Overall Architecture

As illustrated in Figure 2, we propose SPMTrack, a novel tracker that integrates multi-frame spatio-temporal context and employs TMoE for parameter-efficient fine-tuning. The overall architecture follows the one-stream paradigm, consisting of a Feature Extraction Network based on a Vision Transformer [12] as backbone, and a prediction head.

Feature Extraction Network. The feature extraction network is denoted as Φ. The main body of the feature extraction network is a Transformer encoder that utilizes TMoEBlock as its primary component, which is designed based on TMoE. The architecture of TMoEBlock is illustrated in the right panel of Figure 2, and details will be introduced in Section 3.2. The encoder simultaneously performs feature extraction and relation modeling for input images. We extend the input from one single template $\boldsymbol{Z} \in \mathbb{R}^{H_z \times W_z \times 3}$ that is commonly used in one-stream trackers to the video sequence $\{\boldsymbol{Z}_1, ..., \boldsymbol{Z}_N | \boldsymbol{Z}_i \in \mathbb{R}^{H_z \times W_z \times 3}\}$ as reference frames. The reference frames contain the first frame template and the selected tracked historical frames, with a total of N frames. The feature extraction network uses patch embedding to partition each input image into patches of size $M \times M$, map all patches into tokens by a convolutional layer, and obtain the reference token sequence of each image $\boldsymbol{T}_i \in \mathbb{R}^{N_T \times d}$, where d denotes the hidden dimension of Transformer encoder and $N_T = \frac{H_z W_z}{M^2}$ represents the number of tokens corresponding to each image. Similarly, for the search region $\boldsymbol{S} \in \mathbb{R}^{H_s \times W_s \times 3}$, we can also obtain the token sequence $\boldsymbol{X} \in \mathbb{R}^{N_X \times d}$. All tokens are added with positional embedding and a learnable token type embedding [33]. Inspired by [32, 61], we additionally introduce a target state token $\boldsymbol{H} \in \mathbb{R}^{1 \times d}$ that is learnable as well and propagates over time. All tokens are concatenated and fed into the Transformer encoder. The process can be formulated as:

$$\begin{aligned} \boldsymbol{T}_i^j &= \begin{cases} \boldsymbol{T}_i^j + \boldsymbol{PE}_j + \boldsymbol{TE}_o, & if \text{ token } (i,j) \text{ in bbox} \\ \boldsymbol{T}_i^j + \boldsymbol{PE}_j + \boldsymbol{TE}_b, & otherwise \end{cases} \\ \boldsymbol{X}^j &= \boldsymbol{X}^j + \boldsymbol{PE}_j + \boldsymbol{TE}_S \\ \boldsymbol{I} &= \text{Concat}(\boldsymbol{H}, \boldsymbol{T}_1, ..., \boldsymbol{T}_N, \boldsymbol{X}) \end{aligned} \tag{1}$$

where $\boldsymbol{T}_i^j$ represents the jth token in $\boldsymbol{T}_i$, $\boldsymbol{PE}_j$ represents positional embedding of the jth token, similarly for $\boldsymbol{X}^j$. $\boldsymbol{TE}_o$, $\boldsymbol{TE}_b$ and $\boldsymbol{TE}_S$ represents the learnable type embeddings of the foreground region tokens, background region

tokens in reference frames and the search region tokens, respectively. $\boldsymbol{I}$ denotes the input to the Transformer encoder.

Subsequently, the Transformer encoder processes all tokens and produces an output $\boldsymbol{O} \in \mathbb{R}^{(1+N\times N_T+N_X)\times d}$. The output tokens $\boldsymbol{X}' \in \mathbb{R}^{N_X\times d}$ corresponding to search region and the output $\boldsymbol{H}' \in \mathbb{R}^{1\times d}$ from the target state token are fed into the prediction head for prediction. $\boldsymbol{H}'$ will be added with the input target state token in the next tracking frame, which can be formulated as:

$$\boldsymbol{O} = \Phi(\boldsymbol{I}), \ \ [\boldsymbol{H}', \boldsymbol{T}_1', ..., \boldsymbol{T}_N', \boldsymbol{X}'] = \text{Split}(\boldsymbol{O}) \tag{2}$$

Prediction Head. The input to the prediction head consists of the output target state token $\boldsymbol{H}'$ and the output search region tokens $\boldsymbol{X}'$ from the feature extraction network. Initially, the prediction head performs matrix multiplication between $\boldsymbol{H}'$ and $\boldsymbol{X}'$ to obtain the weight $\boldsymbol{U} \in \mathbb{R}^{N_X\times 1}$. As the target state token propagates over time, $\boldsymbol{U}$ can reweight and adjust current search region tokens based on the historical state of the target, thereby introducing more spatio-temporal tracking context. After element-wise multiplication of $\boldsymbol{X}'$ with $\boldsymbol{U}$, the resulting data is reshaped into a feature map $\boldsymbol{F} \in \mathbb{R}^{\frac{H_s}{M}\times\frac{W_s}{M}\times d}$, which is then processed by a decoupled double-MLP head to perform target center classification and box regression, respectively. For fair comparison, the double-MLP head is identical to [33].

3.2. The Design of TMoEBlock

TMoEBlock serves as the primary component of the Transformer encoder in the feature extraction network. As illustrated in the right of Figure 2, TMoEBlock is built based on the standard vision Transformer block. However, unlike the standard ViT block, TMoEBlock applies TMoE module to both the multi-head self-attention (MSA) layer and the feed-forward network (FFN) layer, obtaining MoE-based MSA (M^2SA) layer and MoE-based FFN (MFFN) layer. The finer-grained application of TMoE is capable to achieve more diverse combinations for various relation modeling. The processing of TMoEBlock can be formulated as:

$$\begin{aligned} \boldsymbol{O}_l' &= \text{M}^2\text{SA}(\text{LN}(\boldsymbol{O}_{l-1})) + \boldsymbol{O}_{l-1}, \quad l = 1, ..., L \\ \boldsymbol{O}_l &= \text{MFFN}(\text{LN}(\boldsymbol{O}_l')) + \boldsymbol{O}_l', \quad l = 1, ..., L \end{aligned} \tag{3}$$

where L denotes the total number of TMoEBlocks, $\boldsymbol{O}_l$ represents the output of the lth block in Transformer encoder. When $l = L$, $\boldsymbol{O}_l$ is the output $\boldsymbol{O}$ of the feature extraction network; when $l = 0$, it is the overall input $\boldsymbol{I}$. LN stands for the layer normalization operation.

MoE-based Multi-head Self-Attention (M^2SA). M^2SA not only replaces the standard three linear projection layers $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}$ in standard attention for the query, key, and value with TMoE modules but also replaces the output projection layer with TMoE module.

MoE-based Feed-Forward Network (MFFN). Conventional feed-forward network consists of two linear layer and activation function. MFFN replaces both linear layers in FFN with TMoE modules, while maintaining all other architectural configurations unchanged.

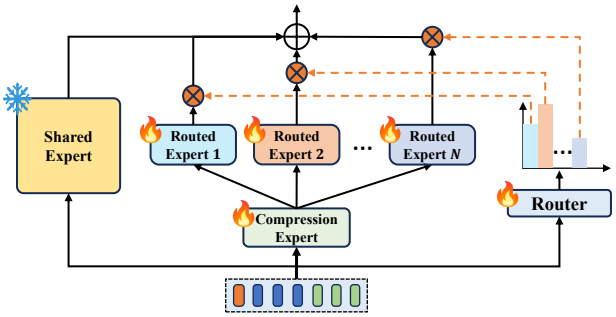


Figure 3. The structure of TMoE. The symbols maintain the same meaning with Figure 2.

3.3. Mixture of Experts for Visual Tracking

In this section, we introduce TMoE, a mixture of experts module tailored for visual tracking task. As illustrated in Figure 3, TMoE consists of one shared expert, one router, one compression expert and N_e routed experts. In large language models, mixture of experts are commonly applied to replace FFN layers, and the experts comprising MoE are typically the same structure as FFN. However, TMoE is used to replace the linear layers in both self-attention and FFN, and the shared expert, the compression expert and the routed experts are all implemented as linear layers.

In TMoE, the shared expert directly copies weights from the corresponding linear layers in pretrained model and remains frozen during training. The rationale is that the original weights have been well-trained during the pre-training stage, and the shared expert needs to retain more general knowledge. In contrast, the compression experts, routed experts, and the router are all trainable.

Given a TMoE module replacing a linear layer with input dimension d and output dimension D, for each token $\boldsymbol{x} \in \mathbb{R}^d$, as shown in Figure 3, TMoE first computes the weights $\boldsymbol{W} \in \mathbb{R}^{N_e}$ of all routed experts based on the input $\boldsymbol{x}$, the weights are normalized by Softmax. After that, TMoE calculates the output $\boldsymbol{y}^s \in \mathbb{R}^D$ of the shared expert and the outputs $\boldsymbol{y}_i^e \in \mathbb{R}^D$ of all routed experts. In particular, instead of adopting the same structure as the shared expert, inspired by [45], we first employ a compression expert with a $d \times r$ linear transformation to reduce the input dimension from d to r and obtain the compressed result $\boldsymbol{y}^c \in \mathbb{R}^r$, where $r \ll d$ and $r \ll D$. Subsequently, all routed experts use the compressed input $\boldsymbol{y}^c$ to produce their respective outputs $\boldsymbol{y}_i^e \in \mathbb{R}^D$. The process can be formally described as:

$$\begin{aligned} \boldsymbol{W} &= \text{Softmax}(\text{Router}(\boldsymbol{x})), \quad \boldsymbol{y}^s = \boldsymbol{E}^S(\boldsymbol{x}) \\ \boldsymbol{y}^c &= \boldsymbol{E}^C(\boldsymbol{x}), \quad \boldsymbol{y}_i^e = \boldsymbol{E}_i^R(\boldsymbol{y}^c) \quad i = 1, ..., N_e \end{aligned} \tag{4}$$

where Router is also a linear layer that maps the input $\boldsymbol{x}$

to weights $\boldsymbol{W}$, and $\boldsymbol{E}^S$, $\boldsymbol{E}^C$, and $\boldsymbol{E}_i^R$ denote the shared expert, compress expert, and i^{th} routed expert, respectively.

After obtaining the outputs of all experts, we compute a weighted sum of the outputs $\boldsymbol{y}_i^e$ from all routed expert based on $\boldsymbol{W}$ to obtain $\boldsymbol{y}^e \in \mathbb{R}^D$, which means that the routed experts are densely activated in TMoE. Subsequently, $\boldsymbol{y}^e$ is added with the shared expert output $\boldsymbol{y}^s$ to get the final output $\boldsymbol{y} \in \mathbb{R}^D$. The process can be formulated as follows:

$$\boldsymbol{y}^e = \sum_{i=1}^{N_e} \boldsymbol{W}_i \boldsymbol{y}_i^e, \qquad \boldsymbol{y} = \boldsymbol{y}^e + \boldsymbol{y}^s \tag{5}$$

3.4. Loss Function

The outputs of the decoupled double-MLP prediction head are supervised using binary cross-entropy loss for target center classification and Generalized IoU loss [42] for bounding box regression, respectively. Both loss terms are assigned with the weighting coefficients set to 1.

4. Experiments

4.1. Implementation Details

Model settings. We provide three versions of SPMTrack: SPMTrack-B, SPMTrack-L, and SPMTrack-G, utilizing ViT-B [12], ViT-L [12], and ViT-G [39] as backbones respectively. All versions maintain a consistent patch size M of 14 and adopt the pretrained weights from DINOv2 [39]. The resolution of input images are the same across all versions, with reference frames of 196×196 and search region of 378×378. A crop factor of 2 is applied to reference frames and 5 to search region. We set the number of reference frames N to 3 for spatio-temporal modeling. Each TMoE module incorporates 4 routed experts, where the input dimension of the routed experts r is set to 64. Other configurations and model information are shown in Table 1, where L is the number of TMoEBlocks, d denotes hidden dimension and N_h is the number of attention heads.

Table 1. Model configurations and parameter counts across SPMTrack variants of different scales.

	SPMTrack-B	SPMTrack-L	SPMTrack-G
Backbone	$L = 12$ $d = 768$ $N_h = 12$	$L = 24$ $d = 1024$ $N_h = 16$	$L = 40$ $d = 1536$ $N_h = 24$
#Params (M)	115.3	379.6	1339.5
#Trainable Params (M)	29.2	75.9	204.0

Datasets. Following previous works [3, 33, 51, 59], we use LaSOT [14], TrackingNet [47], GOT-10K [25], and COCO [34] for training. When evaluating on GOT-10K *test* set, we strictly follow the protocol by training exclusively on GOT-10K *training* set. When evaluating on other datasets, we jointly train on the *training* sets of all four datasets, randomly sampling from each dataset with equal probability. For video datasets, we sample 5 frames from one video with random frame intervals ranging from 1 to 200. The sampled frames are arranged either in forward or reverse order with equal probability, where the first 3 frames serve as reference frames and the remaining 2 as search frames. For COCO, we replicate each image 5 times to maintain consistency with the video frame sampling strategy.

Training and Optimization. Our method is implemented based on PyTorch 2.3.1 and trained on 8 NVIDIA A100 GPUs. We maintain a global batch size of 128 during training. When training exclusively on GOT-10K, we train for 100 epochs. When joint training on multiple datasets, we extend to 170 epochs. In each epoch, we sample 131,072 sequences. We employ AdamW [35] as optimizer with learning rate of 10^{-4} and weight decay of 0.1. The learning rate scheduler warms up over the initial 2 epochs from 10^{-7} to 10^{-4} and then follows a cosine schedule to 10^{-6}.

Inference. We use a total of 3 reference frames, including the annotated first frame template. When current search frame index $t \leq 3$, we use all available tracked frames as reference frames, if there are fewer than three frames, we repeat the template as necessary to reach the required number. When $t > 3$, we select two tracked frames as reference frames at indexes $\lfloor \frac{t}{3} \rfloor$ and $\lfloor \frac{2t}{3} \rfloor$. Following previous trackers [4, 59, 61], we apply a Hanning window penalty to the output response map of the classification head.

4.2. Comparisons with the State-of-the-Art

LaSOT [14] is a large-scale long-term tracking dataset with its *test* set containing 280 sequences. As shown in Table 2, SPMTrack-B significantly outperforms all trackers utilizing ViT-B as backbone. Our method also substantially surpasses LoRAT [33] that also employs parameter-efficient fine-tuning. Additionally, as shown in Figure 4, we evaluate tracker performance across various challenging scenarios in LaSOT. Our method surpasses previous trackers that also incorporate spatio-temporal context [2, 3, 6, 51, 56], while significantly outperforming ROMTrack [4] and GRM [20], which focus on optimizing relation modeling.

GOT-10K [25] contains 9,335 sequences for training and 180 sequences for testing. We follow the protocol that trackers are only allowed to be trained using GOT-10K *train* split. Our approach significantly outperforms LoRAT [33] with the same backbone. And our SPMTrack-B rivals the performance of the current state-of-the-art methods, while SPMTrack-L substantially surpasses all existing trackers.

TrackingNet [38] is a large-scale visual tracking dataset with a *test* set comprising 511 video sequences. Table 2 demonstrates that our method achieves state-of-the-art performance using only ViT-B as backbone, surpassing current trackers that utilize ViT-L or even ViT-G.

Table 2. State-of-the-art comparison on LaSOT, GOT-10k and TrackingNet. '*' denotes for trackers trained only with GOT-10k *train* split. The best three results are highlighted in **red**, **blue** and **bold**, respectively.

Method	Source	LaSOT			GOT-10k*			TrackingNet		
		AUC(%)	P_{Norm}(%)	P(%)	AO(%)	$SR_{0.5}$(%)	$SR_{0.75}$(%)	AUC(%)	P_{Norm}(%)	P(%)
SPMTrack-B	**Ours**	74.9	84.0	81.7	76.5	85.9	76.3	**86.1**	90.2	85.6
SPMTrack-L	**Ours**	**76.8**	**85.9**	**84.0**	**80.0**	**89.4**	**79.9**	**86.9**	**91.0**	**87.2**
SPMTrack-G	**Ours**	**77.4**	**86.6**	**85.0**	**81.0**	**89.2**	**82.3**	**87.3**	**91.4**	**88.1**
LoRAT-B$_{378}$ [33]	ECCV24	72.9	81.9	79.1	73.7	82.6	72.9	84.2	88.4	83.0
LoRAT-L$_{378}$ [33]	ECCV24	75.1	84.1	82.0	77.5	86.2	78.1	85.6	89.7	85.4
LoRAT-G$_{378}$ [33]	ECCV24	**76.2**	**85.3**	**83.5**	78.9	87.8	**80.7**	86.0	90.2	86.1
AQATrack$_{384}$ [56]	CVPR24	72.7	82.9	80.2	76.0	85.2	74.9	84.8	89.3	84.3
ARTrackV2-B$_{384}$ [2]	CVPR24	73.0	82.0	79.6	77.5	86.0	75.5	85.7	89.8	85.5
ARTrackV2-L$_{384}$ [2]	CVPR24	73.6	82.8	81.1	**79.5**	87.8	79.6	86.1	**90.4**	**86.2**
HIPTrack [3]	CVPR24	72.7	82.9	79.5	77.4	**88.0**	74.5	84.5	89.1	83.8
ODTrack-B [61]	AAAI24	73.2	83.2	80.6	77.0	87.9	75.1	85.1	90.1	84.9
F-BDMTrack-384 [58]	ICCV23	72.0	81.5	77.7	75.4	84.3	72.9	84.5	89.0	84.0
ROMTrack-384 [4]	ICCV23	71.4	81.4	78.2	74.2	84.3	72.4	84.1	89.0	83.7
ARTrack$_{384}$ [51]	CVPR23	72.6	81.7	79.1	75.5	84.3	74.3	85.1	89.1	84.8
SeqTrack-B$_{384}$ [6]	CVPR23	71.5	81.1	77.8	74.5	84.3	71.4	83.9	88.8	83.6
GRM [20]	CVPR23	69.9	79.3	75.8	73.4	82.9	70.4	84.0	88.7	83.3
OSTrack$_{384}$ [59]	ECCV22	71.1	81.1	77.6	73.7	83.2	70.8	83.9	88.5	83.2
SimTrack [5]	ECCV22	70.5	79.7	-	69.8	78.8	66.0	83.4	87.4	-

TNL2K [49] comprises 700 test video sequences, each accompanied by natural language descriptions. As shown in Table 3, our method achieves state-of-the-art performance without utilizing natural language descriptions and surpasses existing vision-language trackers [21, 30].

UAV123 [37] is a dataset focusing on low-altitude drone aerial scenarios, consisting of 123 video sequences with an average length of 915 frames per sequence. Our method achieves state-of-the-art performance with comparable model sizes, demonstrating the capability of our approach in aerial scenes and small object tracking.

NfS [29] consists of 100 video sequences totaling 380,000 frames. Following previous works, we conduct evaluations on the 30 FPS version of the dataset. Our method rivals current state-of-the-art trackers in NfS. One possible reason for not achieving further improvement is that our method does not incorporate consecutive target position like ARTrack [51] and HIPTrack [3], which is crucial for NfS.

OTB2015 [54] is a classic dataset in visual tracking, consisting of 100 short-term tracking sequences that encompass 11 common challenges, such as target deformation, occlusion, and scale variation. Our method also achieves state-of-the-art performance on OTB2015.

4.3. Ablation Study

The Importance of Spatio-Temporal Context Modeling and TMoE. Table 5 demonstrates the significance of two core designs in our paper: spatio-temporal context modeling and TMoE. In Table 5, removing TMoE indicates direct fine-tuning with LoRA [24], while removing spatio-temporal modeling indicates using only the first frame template and search region as input and eliminating the target state token. Compared to LoRA, applying TMoE for parameter-efficient fine-tuning yields a substantial improvement of **+0.8** AUC on LaSOT, demonstrates the superiority of TMoE. And the spatio-temporal context modeling further enhances performance by an additional **+1.2** AUC. On the basis of incorporating spatio-temporal modeling, the improvement brought by TMoE is **+1.0** AUC, indicating that TMoE can help the model better capture spatio-temporal contextual information.

Table 3. The performance of our method and other state-of-the-art trackers on TNL2K. The best three results are highlighted in **red**, **blue** and **bold**.

Method	AUC(%)	P_{Norm}(%)	P(%)
SPMTrack-G	**64.7**	**82.6**	**70.6**
SPMTrack-L	**63.7**	**81.5**	**69.2**
SPMTrack-B	**62.0**	**79.7**	**66.7**
LoRAT-B$_{378}$ [33]	59.9	-	63.7
ODTrack-L [61]	61.7	-	-
ARTrackV2-L$_{384}$ [2]	61.6	-	-
ARTrack-L$_{384}$ [51]	60.3	-	-
RTracker-L [26]	60.6	-	63.7
UNINEXT-H [57]	59.3	-	62.8
CiteTracker [30]	57.7	-	59.6
VLT [21]	53.1	-	53.3

Where to Apply TMoE. Conventional MoE is typically applied to the FFN layers within Transformer blocks, while keeping the attention layers unchanged. In contrast, we ap-

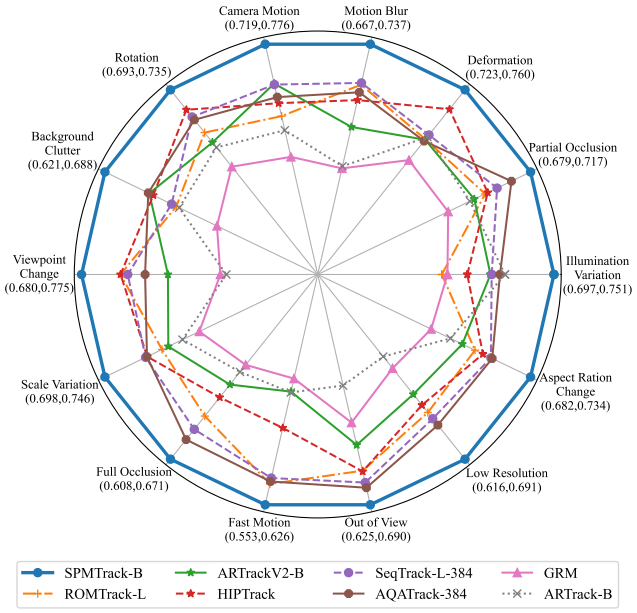


Figure 4. The performance of our method compared with other state-of-the-art trackers in terms of AUC across various scenarios in the LaSOT *test* split.

Table 4. The performance of our method and other state-of-the-art trackers on UAV123, NfS and OTB2015 in terms of AUC metrics. The best three results are highlighted in **red**, **blue** and **bold**.

Method	UAV123	NfS	OTB2015
SPMTrack-B	**71.7**	**67.4**	**72.7**
HIPTrack [3]	**70.5**	**68.1**	71.0
ARTrackV2-B [2]	69.9	**67.6**	-
ODTrack-L [61]	-	-	**72.4**
ARTrack_{384} [51]	**70.5**	66.8	-
SeqTrack-B_{384} [6]	68.6	66.7	-
DropTrack [52]	-	-	69.6
MixFormer-L [10]	69.5	-	-
STMTrack [16]	64.7	-	**71.9**

Table 5. Ablation studies on spatio-temporal context modeling and TMoE. Experiments are conducted on LaSOT.

#	Spatio-Temporal	TMoE	AUC (%)	P_{Norm}(%)	P(%)
1	✘	✘	72.9	81.9	79.1
2	✘	✔	73.7	82.7	80.0
3	✔	✘	73.9	82.8	80.0
4	✔	✔	**74.9**	**84.0**	**81.7**

ply TMoE in both the attention layers and the FFN layers. Table 6 demonstrates the impact of applying TMoE in different layers. Comparing the first, third, and fifth rows, applying TMoE to the query, key, value projections in attention layers and to FFN layers can both improve performance. Comparing the first, second and third rows, we find that applying TMoE in attention layers yields significantly greater benefits compared to applying it only in FFN layers. Comparing the third and fourth rows, applying TMoE in the output projection matrix of the attention layer can bring a slight improvement.

Table 6. Ablation studies on applying TMoE in different layers. All results are evaluated on LaSOT *test* split. "✘" means replacing TMoE with LoRA.

#	QKV	Out Proj	FFN	AUC (%)	P_{Norm}(%)	P(%)
1	✘	✘	✘	73.9	82.8	80.0
2	✘	✘	✔	74.1	83.1	80.8
3	✔	✘	✘	74.5	83.6	81.2
4	✔	✔	✘	74.6	83.8	81.4
5	✔	✔	✔	**74.9**	**84.0**	**81.7**

Whether to use a shared compress expert. In TMoE, we employ a shared compression expert to reduce the input dimension to r, and use multiple routed experts for subsequent processing. As shown in the first and second rows of Table 7, compared to providing each routed expert with a compression expert, sharing the compression expert can achieve better performance while reducing the parameter count. When not sharing the compression expert, we provide t-SNE visualizations of the outputs from compression experts and routed experts in five TMoE modules in Figure 5. The outputs of different compression experts show significant overlaps, while the routed experts are more dispersed, validating the effectiveness of our approach.

TMoE vs. Conventional MoE. We compare TMoE with conventional MoE on the visual tracking task. For conventional MoE, we use LoRA to fine-tune the attention layers and apply MoE in the FFN layers with five experts: one frozen shared expert that copies pretrained FFN parameters, and four trainable routed experts, all routed experts are identical to the original FFN architecture, and a router is also applied. As shown in the first row and third row of Table 7, TMoE demonstrates significant advantages over conventional MoE in both parameter efficiency and performance.

Table 7. Ablation studies on whether to share the compression expert and use the conventional MoE to replace TMoE. Results are evaluated on LaSOT *test* split.

model variants	#Params(M)	AUC (%)	P_{Norm}(%)	P(%)
SPMTrack-B	**115.3**	**74.9**	**84.0**	**81.7**
w/o shared compress expert	131.3	74.8	83.7	81.5
Conventional MoE	316.8	73.4	82.6	79.8

The Number of Routed Experts. In Table 8, we evaluated the impact of different numbers of routed experts in TMoE. Results indicate that increasing the number of experts leads to improved performance, highlighting the potential of TMoE as a method to expand the parameter count and model scale. To balance the number of parameters and

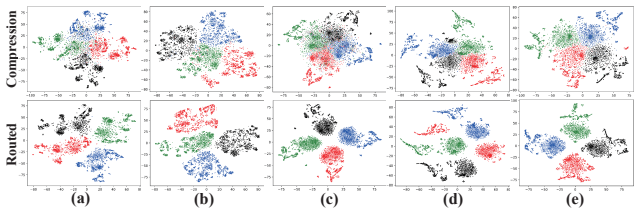


Figure 5. Comparison of t-SNE visualizations. Each column shows outputs from all compression experts (top) and routed experts (bottom) within a TMoE module. Different colors represent distinct experts. Zoom in for better view.

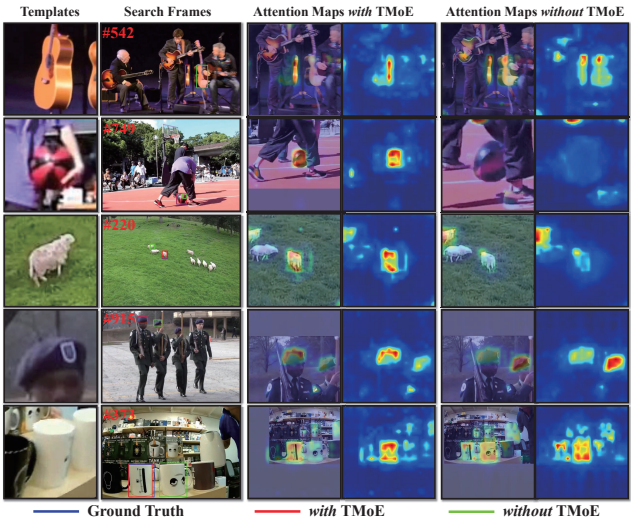


Figure 6. Visualization comparison of search region attention maps *with* and *without* TMoE. Zoom in for better view.

performance, we choose to use 4 experts, and whether there are separate optimal configurations for the TMoE in the attention layer and the FFN layer remains to be explored.

The Number of Reference Frames. In SPMTrack, we use the template and several tracked frames as reference frames to enhance the spatio-temporal modeling ability. In Table 9, we explore the impact of using different numbers of reference frames during training and inference. During inference, if only two reference frames are used, we simply select the frame at $\lfloor \frac{t}{2} \rfloor$ as the reference frame. If four reference frames are used, we select the frames at $\lfloor \frac{t}{4} \rfloor$, $\lfloor \frac{t}{2} \rfloor$, and $\lfloor \frac{3\times t}{4} \rfloor$. As shown in Table 9, better performance can be achieved when the number of reference frames used during inference remains the same as that during training. Training with more reference frames can also lead to improved performance, illustrating that our method still has substantial potential for performance enhancement.

Comparison of attention maps with and without TMoE. In Figure 6, to exclude the influence of intermediate tracked frames, we use the model in rows 1 and 2 of Table 5 for comparison, where spatio-temporal modeling is removed and we focus solely on the impact of TMoE. We compare the attention maps of the search region in the last block of Transformer encoder. The first column is the template, the second column displays search region, the third column is the visualization result and pure attention map, as well as the fourth column. Figure 6 shows that when facing challenging scenarios such as occlusions, distractors, and viewpoint variations, TMoE effectively suppresses background regions while enhancing perception of the target boundary.

Table 8. The performance of our method on the *test* split of GOT-10K when setting different number of routed experts in TMoE.

Number	2	4	6	8
AO(%)	75.3	76.5	76.6	**77.1**
$\mathbf{SR}_{0.5}$(%)	84.4	85.9	86.3	**86.4**
$\mathbf{SR}_{0.75}$(%)	75.2	76.3	77.0	**77.6**

Table 9. Ablation study on the number of reference frames during training and inference. Results are evaluated based on SPMTrack-B on GOT-10K *test* split.

Training	Inference	AO (%)	$\mathbf{SR}_{0.5}$(%)	$\mathbf{SR}_{0.75}$(%)
2	2	**75.8**	**85.1**	**75.3**
	3	74.7	84.6	73.8
	4	69.8	81.8	66.2
3	2	73.1	83.9	72.7
	3	**76.5**	**85.9**	**76.3**
	4	72.8	83.4	71.3
4	2	70.6	83.9	68.9
	3	74.6	85.4	74.5
	4	**77.5**	**87.3**	**77.2**

5. Conclusion

In this paper, we present TMoE, a mixture of experts module tailored for visual tracking, and propose SPMTrack, a novel tracker enabling spatio-temporal context modeling based on TMoE. TMoE is applied in the linear layers in both self-attention and FFN layers, enhancing the diversity and flexibility of expert combinations to better handle various relation modeling in visual tracking. Additionally, TMoE employs a lightweight and efficient structure and serves as a method of parameter-efficient fine-tuning, which enables us to train SPMTrack of larger scales and enables SPMTrack to achieve state-of-the-art performance with only a small subset of parameters need to be trained. Furthermore, we hope that this work will inspire more applications of mixture of experts in the field of visual tracking.

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